

Rhythms of Power: Politics and Temporal Justice

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1 Juillet, 2026

Standalone article extending the treatise *Time May Not Exist!* (Brécheteau, 2025)

DOI: 10.5281/zenodo.17214502

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Abstract

This article explores the entanglement of power, technology and temporality through the lens of the *Chronon Field*, $\Phi(x)$, introduced in the treatise *Le temps n'existe peut-être pas !*. It argues that the contemporary political crisis is rhythmic as much as institutional: a loss of coherence under the acceleration imposed by digital infrastructures. By articulating physical time, social synchronization and the thermodynamics of coherence, it formalizes the notion of *temporal justice* via operational indicators (rhythmic budget $B(H)$, *Temporal Access Gap*, *Worst-Case Access*, uniformity index $U(H)$, *basic rhythmic credit*), and proposes an empirical and normative program for a *politics of tempo* compatible with the constraints of living systems.

Keywords: Temporal justice, Chronon Field, synchronization, technological power, rhythmic coherence, public time (UTC/GNSS), social temporality

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I — Governing time

Each era imposes its own tempo. In the nineteenth century, factories set the day to the beat of the machines: the morning siren sounded at 5:45 a.m., marking entry into an industrial time where steam dictated the cadence. In the twentieth, radio, transport and television unified distant spaces to the nearest second, tying lives to the rhythm of a global clock. In the twenty-first, power has changed form: it no longer only imposes laws, it sketches rhythms.

Late modernity is not just an era of speed; it is an era of *time capture*. Territories matter less than flows: financial, digital, cognitive. Algorithmic economy, networks and high-frequency markets shape a normative temporality in which everyone must remain “synchronized”. A latency of 10^{-3} s already excludes an order in HFT; a few seconds are enough to render sensitive information obsolete. Failing to answer in time is already to disappear.

But time is not a private resource: it is a *commons*. When some accelerate while others stagnate, a new fracture opens up — that of *tempo inequalities*.

II — The tempo divide

Globalization has unified the *measurement* of the world, not its *cadence*. Optical fibers link Tokyo to New York in ~ 60 ms, while a local public service faces latencies and queues one hundred to one thousand times larger. This asymmetry is no longer merely economic but rhythmic: some live at the microsecond, others at the minute.

Hyper-connected elites inhabit an anticipated, algorithmic time where the present is already modeled in data. Precarious workers, under-resourced institutions, peripheral territories move in a time of waiting and saturation. Some surf on the instant; others drown in it. Some enjoy an “elastic time” — delegable to machines — while others endure a constrained time, with no breathing space. A market operator

acts on 10^{-6} s; a nurse on 10^3 s. This factor of a billion between their cadences generates an *ontological desynchronization*.

Thus arises a form of *temporal injustice*: unequal access to slowness, rest and deceleration. And physics joins politics here: if reality holds together through coherence, a society deranged in its tempos falls out of tune with the world it claims to inhabit.

III — Machines that beat faster than we do

Our tools have outrun our rhythm. Processors operate at 10^9 operations per second; financial networks trade on microsecond scales; humans integrate a piece of information in ~ 200 ms. Between these orders of magnitude, an abyss has opened: the technical beat has become autonomous, turning into a *flux of interruptions* that starves long phases of integration.

In this regime, slowness is no longer a virtue but a fault. The present fragments into micro-emergencies: notifications, alerts, reminders. We live in a world that restarts faster than it understands itself. Yet no system can sustain itself without phases of rest: sleep–wake cycles, circadian rhythms, attentional windows; right up to stars that pulse with stable periods. Abolishing these breaths means imposing a saturated $\Phi(x)$, *without silence*, which jeopardizes the coherence of living systems.

IV — The Chronon Field and the misaligned society

The Chronon Field $\Phi(x)$ — local cadence of coherence — is not a force but a *frequency* that makes synchronization possible. Two consequences follow. First, $\Phi(x)$ admits *gradients*: $\nabla\Phi(x)$ expresses rhythm differentials between places, devices, collectives. Second, these gradients are *metric*: optical clocks compare relative frequencies at the 10^{-18} level, while UTC/GNSS, PTP or NTP chains already instrument society on μ s–ms scales. Thus the same grammar — phases, frequencies, latencies — describes both **technical** infrastructures and **social** environments.

Collective harmony requires that diverse frequencies tune to one another without merging. Yet modernity de facto imposes a uniform $\Phi(x)$ — a global digital tempo — ignoring the plurality of human beats. The result is a *structural desynchronization*: some are exhausted, others are bored, everyone is out of step. Rhythm then becomes a principle of alignment: we must reintroduce delays, intervals and rest.

Stability condition

$$\text{Var}_H[\Phi_g] \leq \Delta\Phi_c^2 \quad \text{and} \quad (\text{TAG}, U(H)) \searrow \text{ or constant while } B_{\min} \nearrow$$

No policy should *simultaneously* increase inequalities of access (TAG) and uniformization $U(H)$ while lowering the floor $B_{\min}$.

Box — Metric toolbox: $B(H)$, **TAG**, **WCA**, $U(H)$ **Framework.** Horizon H (day/week/month); τ = metric time (UTC/TAI); Φ in s^{-1} .

Rhythmic budget $B(H)$.

$$B(H) = \int_H \Phi(x) d\tau \quad (\text{dimensionless})$$

Interpretation: cumulative “rhythmic opportunity” (number of coherence cycles). *Quantiles.* B_q (global) and $B_q^{(g)}$ (group g).

Access inequalities (TAG).

$$\text{TAG} = B_{90} - B_{10} \quad (\text{same unit as } B)$$

Small TAG = more homogeneous access to long phases (rest/attention).

Worst-Case Access (WCA).

$$\text{WCA} = W_{0.99} \quad (\text{seconds or minutes})$$

$W_{0.99}$ = 99th percentile of access delays to a critical service (emergency triage, counter, support ticket).

Uniformization $U(H)$.

$$U(H) = \frac{\sum_{f \in \mathcal{F}_{\text{forced}}} \tau_f}{\sum_{f \in \mathcal{F}} \tau_f} \quad (\text{dimensionless, } 0 \rightarrow 1)$$

Share of *forced* flows in the same cadence window (single profile). Aim for a low $U(H)$.

Floor $B_{\min}$. Enforceable threshold on $B(H)$ guaranteeing rest and *human delays* (24–48 h outside emergencies); calibrated on *outcomes* (quality, safety).

Numerical examples (illustrative).

- *Networks (user support, $H = 1$ week).* $W_{0.99} = 180$ s; $U(H) = 0.72$ (peaks 9–12 a.m. & 2–5 p.m.); after spreading time slots: $U(H) = 0.49$, TAG $\downarrow$ by 22%.
- *Transport (commuting).* $B_{10} = 8.1 \times 10^4$, $B_{90} = 1.2 \times 10^5 \Rightarrow \text{TAG} = 3.9 \times 10^4$; pricing & staggered schedules $\Rightarrow$ TAG $\downarrow$ by 30%.
- *Hospitals (access to non-urgent imaging).* $W_{0.99} = 21$ days; introduction of “low-frequency” slots (evening/weekend): $W_{0.99} = 12$ days, $B_{\min} \uparrow$.

Epistemic status The social extension of $\Phi(x)$ is a *structural analogy*: same operators (phase, frequency, coherence), shared metrics (latency, *jitter*), falsifiability criteria (latency audits, dispersion of B_q by group, response to delay policies). It does not reify society as a “field”; it proposes a *common language* linking metrology and institutions, testable through audit and intervention protocols (preregistration, controls, multi-site replication).

V — Politics of tempo

To demand temporal justice is not to reject progress; it is to demand a fair distribution of the *right to cadence*. Power in the twenty-first century is temporal: it decides who is allowed to slow down. The central question is no longer only “what to produce” or “how to share”, but “*at what speed to live*”.

A politics of tempo defines *zones of compatible frequency*. Some spaces of acceleration exist and are desirable (research, simulation, emergency response); others require slowness (healthcare, education, justice, creation). We may speak of a *rhythmic federalism*: each sphere preserves its own tempo while remaining tuned to the collective beat. Progress would no longer be measured by raw speed, but by the *accuracy of the common beat* — the phase coherence between local $\Phi(x)$.

Empirical and normative program 1) *Latency and jitter audit* of public and critical services (PTP/NTP, actual queues), publication of B_q and $U(H)$ curves by sector and group. 2) *Human delay floors* integrated into indicators: 24–48 h outside emergencies, with evaluation based on *outcomes* (quality, safety), not speed alone. 3) *Minimal multi-standard* requirement: at least two cadence profiles certified as *outcome-equivalent* in each service (to prevent uniformization). 4) *Basic rhythmic credit*: weekly guarantee of rest and deep attention measured by $B(H)$ and verified by random audits. 5) *Rhythmic Pareto rule*: no reform should increase TAG and $U(H)$ simultaneously while decreasing $B_{\min}$ (cf. stability condition).

Use cases (replayed/simulated) & reproducibility

- **Networks.** Replay of incident logs (anonymized), measurement of $W_{0.99}$ and $U(H)$, experimentation with asynchronous windows; protocol: horizon $H = 4$ weeks, $N \geq 10^5$ tickets, preregistered thresholds.
- **Transport.** Entry/exit data (open data), estimation of B_q by municipality and $U(H)$ by line; A/B test of staggered timetables; outcome criterion: TAG reduction $\geq 20\%$ without increase in $W_{0.99}$.
- **Hospitals.** Waiting lists by department, measurement of $W_{0.99}$ and patient $B_{\min}$; “low-frequency” pilot slots (evening/weekend); success if $W_{0.99} \downarrow$ and $B_{\min} \uparrow$ at *constant quality*.

VI — Towards an ethics of slowness and synchronization

The future does not belong to the fastest, but to the most *in tune*. A sustainable civilization is not built on racing, but on measuring the breath. Slowness is not inertia: it is power of resonance, the capacity to come into phase with the real.

Rehabilitating slowness means giving the world back its shared respiration. It means allowing technology, economy and consciousness to recover a stable *phasing* with the field $\Phi(x)$. Biology and physics alike remind us that coherence emerges within specific frequency bands: neither too fast nor too slow. Likewise, society sustains itself by respecting the proper amplitude of lived time.

The Chronon Field, $\Phi(x)$, reminds us that any system — physical or social — survives only by respecting its rhythm of coherence. Temporal justice is not a utopia: it is a condition of reality.

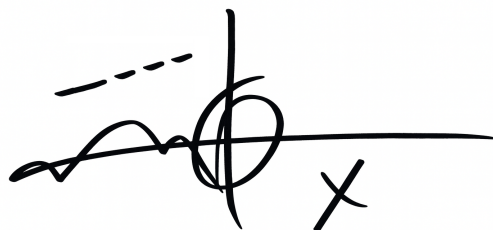
Where rhythms break, worlds collapse. Where they tune, the present becomes habitable.

Box — Ethical principles (transparency, limits, non-stigmatization)

- **Transparency.** Published metrics (B_q , TAG, WCA, $U(H)$), preregistered methods and thresholds; public changelog of rule updates.
- **Data minimization.** Aggregated latencies and queues; irreversible anonymization; no inference of sensitive attributes.
- **Non-stigmatization.** Indicators serve *provisioning* (time slots, resources), never the penalization of a group; fairness audits (differences in $B_q^{(g)}$ under outcome control).
- **Scientific robustness.** Multi-site replication, controls, preregistration; safeguards against Goodhart’s law (indicator rotation, independent reviews).
- **Right to explanation & redress.** Any automated decision on cadences entails a right to explanation, correction, and human appeal.
- **Limits.** Correlation $\neq$ causation; metrics do not capture all social value (care, pedagogy, creation). Final decisions are *contextualized*.

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